

ROUTEFLOW

Climbing Route Management System - MVP Case Study

Role Full-Stack Developer	Project type Self-initiated MVP	Focus Business-oriented web application
------------------------------	------------------------------------	--

The problem

RouteFlow is designed for climbing gyms that need a central place to manage climbing routes, route setters and the lifecycle of routes on the wall.

The MVP

The current version establishes the core full-stack architecture and the foundation for route management. The project is intentionally presented as an MVP and is still under active development.

Technology

Backend: Python, Django, Django REST Framework
Frontend: React, TypeScript, Tailwind CSS
Database: PostgreSQL
Environment: Docker, Docker Compose
Workflow: Git, GitHub

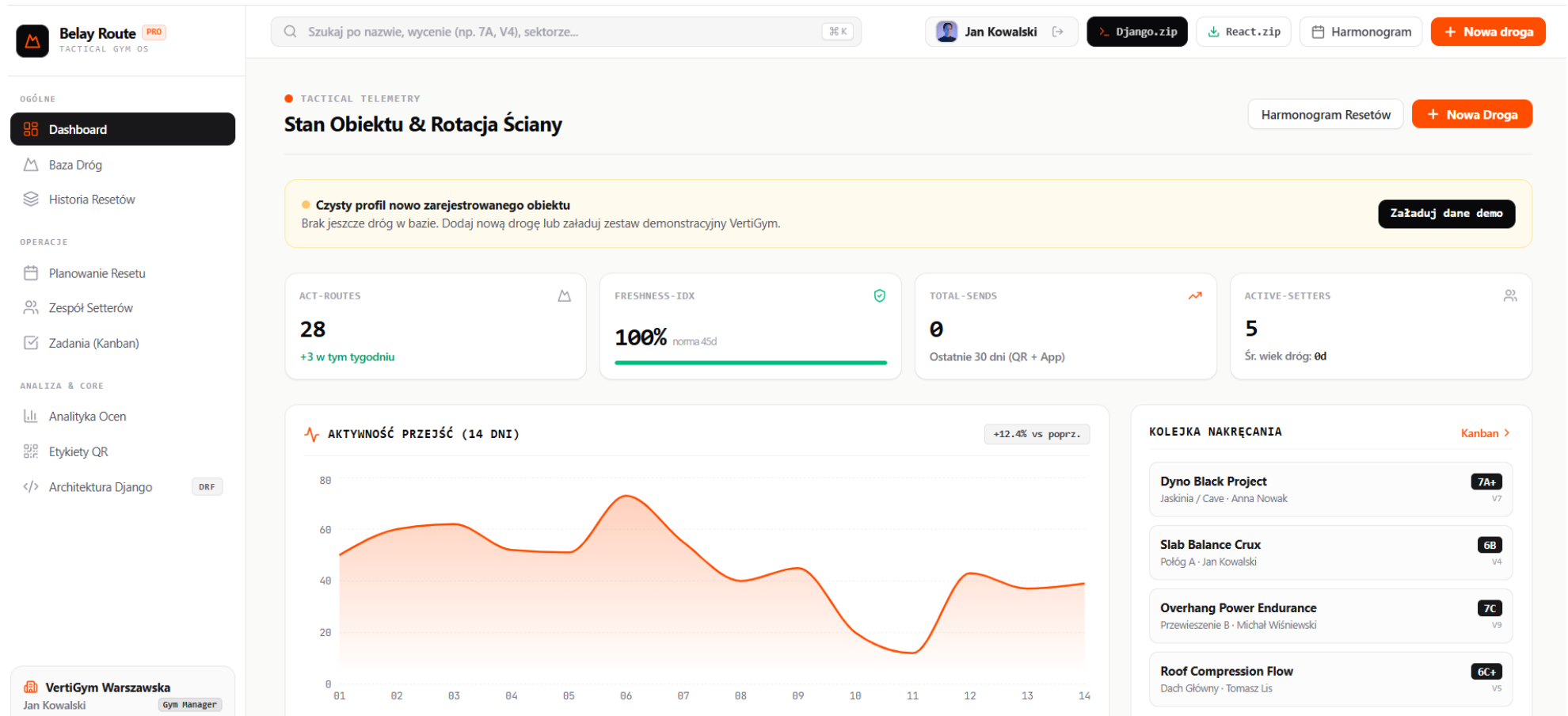
Current scope

Route data, grades, colors, disciplines, setters and route lifecycle information, exposed through a REST API and consumed by the frontend.

DJANGO REST FRAMEWORK	REACT + TYPESCRIPT	POSTGRESQL	DOCKER
-----------------------	--------------------	------------	--------

CURRENT USER INTERFACE

Actual RouteFlow frontend - dashboard view



The dashboard is designed around climbing-gym operations: route inventory, reset history and planning, setter workflow, analytics and QR-related functionality. Some dashboard elements represent the planned product direction while the underlying MVP is still being developed.

APPLICATION ARCHITECTURE

A separated frontend and backend connected through a REST API



REST API development Django REST Framework endpoints and serializers.	Relational data Django ORM and PostgreSQL for application data.
Frontend integration React/TypeScript client consuming backend data.	Containerized environment Docker Compose for application and database services.

Development roadmap

Route-setting planning • setter assignments • reset history • route-age tracking • popularity analytics • QR codes • authentication • role-based permissions

Source code: github.com/Efowski/routeflow